Campus Recruit Analyzer (CRA) E-R Model Documentation

1. Introduction

This document presents the Entity-Relationship (E-R) Model for the Campus Recruit Analyzer (CRA)—a Python-based desktop application designed specifically for students of Dr. B.R. Ambedkar National Institute of Technology, Jalandhar.

The E-R model defines all persistent data entities, their attributes, and relationships among them. The design derives from Software Requirements Specification (SRS v1.0) and UML Use Case document (v2.0), ensuring design artifact consistency.

CRA overcomes generic preparation platform limitations by providing company-specific, college-contextualized guidance through historical placement data, AI-driven summarization, and structured progress tracking.

2. Actors

These actors interact with the system and shape the data model:

Actor	Type	Description
Student	Primary	3rd/4th year NITJ students using system for placement preparation.
Administrator	Primary	Faculty coordinators/student developers managing data, companies, LLM cache.
Placement Cell/TPO	Secondary	Stakeholders monitoring anonymized analytics and preparation trends.

3. Core Entities and Attributes

Core entities support all CRA functionality and analytics.

3.1 STUDENT

Represents registered users, storing authentication and profile data.

Attribute	Key	Data Type	Description
student_id	PK	INT	Unique student identifier
nitj_email	UNIQUE	VARCHAR	Institutional email for login

Attribute	Key	Data Type	Description
password_hash	—	VARCHAR	Bcrypt-hashed password
email_cred_enc	—	TEXT	AES-256 encrypted IMAP credentials

3.2 VISIT

Company recruitment visits to NIT Jalandhar, including compensation details.

Attribute	Key	Data Type	Description
visit_id	PK	INT	Unique visit record identifier
company_id	FK	INT	References associated company
ctc_lpa	—	DECIMAL	Compensation (Lakhs Per Annum)
year	—	INT	Recruitment year

3.3 COMPANY

Participating campus recruitment companies.

Attribute	Key	Data Type	Description
company_id	PK	INT	Unique company ID
company_name	—	VARCHAR	Company name
industry_sector	—	VARCHAR	Industry classification

3.4 TOPIC

Placement preparation topics.

Attribute	Key	Data Type	Description
topic_id	PK	INT	Unique topic identifier
topic_name	—	VARCHAR	Topic name
category	—	VARCHAR	Domain (DSA, OS, DBMS, etc.)

3.5 STUDENT_PROGRESS

Tracks preparation progress per topic.

Attribute	Key	Data Type	Description
progress_id	PK	INT	Unique progress ID
student_id	FK	INT	Associated student

topic_id	FK	INT	Associated topic

Attribute	Key	Data Type	Description
is_completed	—	BOOLEAN	Completion status

3.6 LLM_SUMMARY

AI-generated company preparation summaries.

Attribute	Key	Data Type	Description
summary_id	PK	INT	Unique summary ID
visit_id	FK	INT	Associated visit
summary_text	—	TEXT	AI-generated roadmap

3.7 COMPANY_TOPIC

Maps topics to company visits.

Attribute	Key	Data Type	Description
company_topic_id	PK	INT	Unique mapping ID
visit_id	FK	INT	Associated visit
topic_id	FK	INT	Associated topic

3.8 ANALYTICS_EVENT

Anonymized user interaction analytics.

Attribute	Key	Data Type	Description
event_id	PK	INT	Unique event identifier
student_id	FK	INT	Nullable for anonymization
event_type	—	VARCHAR	Interaction type

4. Relationship Summary

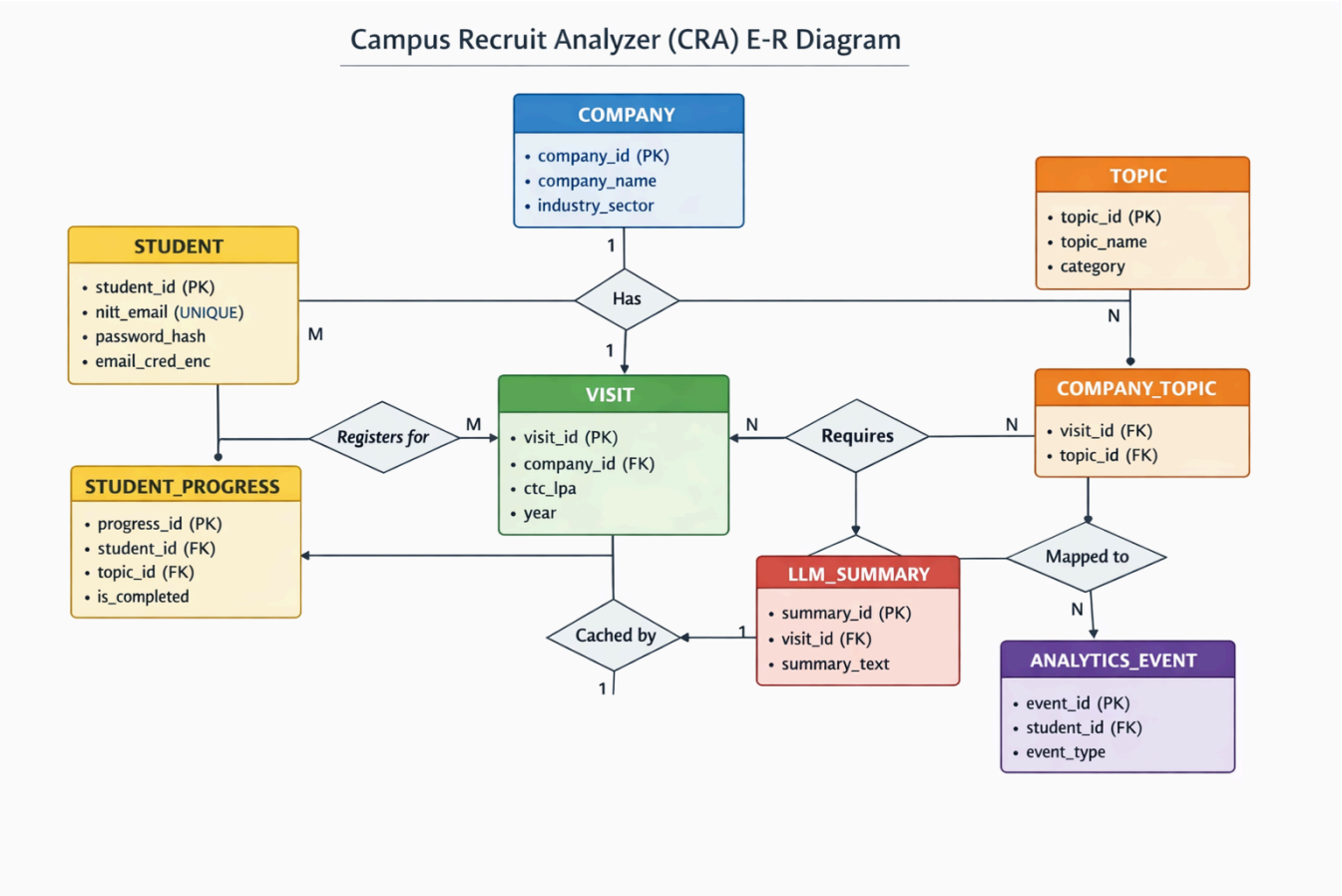
Relationship	Entity A	Cardinality	Entity B	Notes
Registers for	STUDENT	M:N	VISIT	Via STUDENT_PROGRESS
Has	COMPANY	1:N	VISIT	Multiple yearly visits per company
Requires	VISIT	M:N	TOPIC	Via COMPANY_TOPIC
Cached by	VISIT	1:1	LLM_SUMMARY	One summary per visit

5. Traceability to Use Cases

Entity	Use Cases Covered	Purpose
STUDENT	UC-01, UC-02, UC-03	Authentication, profile management
VISIT	UC-05, UC-11	Roadmaps, company data management
STUDENT_PROGRESS	UC-07, UC-08	Progress tracking
LLM_SUMMARY	UC-05, UC-06, UC-14	AI recommendations, cache handling

6. Assumptions and Constraints

Data Persistence: Student progress records use soft-delete (archiving) to preserve history. **Security**: Email credentials stored with AES-256 encryption for confidentiality.

Privacy: Analytics anonymized; no PII exposed in reports/dashboards. **Connectivity**: Requires internet for LLM APIs and SMTP notifications.

System Scope: Non-scalable, single-use, SRS specification.